Machine Learning-Based Prediction of Thermal Conductivity Enhancement in Nanofluids

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***Abstract*—In this paper, we use machine learning techniques to predict the enhancement of the thermal conductivity of nanofluids, which are fluids enhanced with nanoparticles. Nanofluids are used in many modern applications due to their superior heat transfer properties as compared to conventional fluids. However, measuring or determining experimental values of thermal conductivity is both time-consuming and expensive. To accomplish our goal we used a dataset (of nanoparticles and base fluids) containing the thermophysical properties (temperature, particle size, volumetric fraction) needed to predict thermal conductivity. We performed feature engineering through categorical encoding and introduced physically meaningful features (e.g., conductivity ratio - kp/km) to the dataset before training. We evaluated the predictive capabilities of several different regression models, including Linear Regression, Support Vector Regression (SVR), Random Forest, and XGBoost, using the following metrics: Mean Absolute Error (MAE); Root Mean Square Error (RMSE); and R-Squared (R^2). Our results indicated that models based on an ensemble approach outperformed standard (conventional) regression models as well as theoretical correlations that is the Koo–Kleinstreuer–Li (KKL) Model. Therefore, this study demonstrated that machine learning techniques can be successfully used to predict the thermal properties of nanofluids with great accuracy and can substantially reduce the dependence on experimental testing.**

I. Introduction

Nanofluids are suspensions of nanoparticles dispersed within base fluids, where the purpose of creating a nanofluid is to improve the properties of the base fluid for transferring heat. They have a greater conductivity than base fluids and can enhance heat transfer capabilities, making them an attractive option for a variety of applications such as cooling technologies, energy-generating devices, and electronic components.

Accurately predicting the thermal conductivity of a nanofluid is key to optimizing performance within each of these application types. Traditional methods of prediction such as experimental measurements and theoretical correlation techniques (e.g. KKL) often fail to represent the range of nonlinear interactions that can exist between the various influencing factors.

The rise of more advanced machine learning methodologies has resulted in a growing number of data-driven alternatives to traditional prediction methods. Machine learning models can simultaneously analyze many different parameters and can find complex hidden patterns that are difficult or impossible to identify analytically. In this project, we have developed a comparative machine learning framework for predicting thermal conductivity enhancement based on multiple regression models and performing a comparative analysis of those model results when compared to traditional analytical predictions.

II. Methodology

A. Data Collection and Dataset Description

The present study used a set of thermophysical properties of nanofluids which includes both our own data and that which we theorized. This dataset is designed for predicting the thermal conductivity of nanofluids using a robust set of experimental and theoretical parameters. The dataset comprises 278 samples across twelve nanofluid combinations, involving four types of nanoparticles and three base fluids. We looked at key variables which are temperature, particle size, volume fraction, thermal conductivity of nanoparticles (k_p) and that of the base fluid (k_m). We imported and preprocessed this data using the Pandas library for in depth analysis. In this study we use these as input parameters to develop models of nanofluid thermal conductivity. Of all the variables we looked at the experimental thermal conductivity (Exp-TC) which we used as the main focus, also we included in the analysis the theoretical thermal conductivity from the Koo Kleinstreuer Li (KKL) model (KKL-TC) for a comparative look.

B. Data Preprocessing

The data set was looked at for missing and null values at first to guarantee data quality. We found out that there were not to have any large scale missing entries which is to say we did not use imputation techniques.

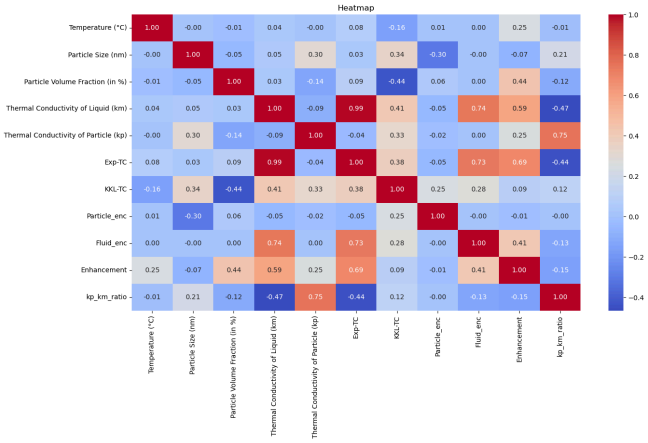
We transformed categorical features like Particle Material and Base Fluid into numbers using the Label Encoding method. In this we put each category into a unique integer which in turn made machine learning algorithms to do their job better. That which is also true is most regression models require numerical input for training.

C. Feature Engineering

Feature engineering was done to make the model work better and add insights from the field. We came up with a new target variable called "thermal conductivity enhancement," which is the difference between the experimental thermal conductivity and the base fluid conductivity. This formulation enables the model to concentrate on the relative enhancements provided by nanoparticles instead of absolute figures. Additionally, a new feature, the conductivity ratio (k_p/k_m) was introduced. This parameter shows how the conductivity of the nanoparticles compares to the conductivity of the base fluid. It also shows how the two interact with each other on a physical level, which helps the model learn how different materials interact with each other.

D. Exploratory Data Analysis

A correlation heatmap was used to look at the relationships between the features as part of exploratory data analysis. The heatmap shows correlation coefficients by using different colour intensities. Values close to +1 mean a strong positive correlation, values close to −1 mean a strong negative correlation, and values close to 0 mean a weak or no correlation. This analysis helps find features that have a big effect on the target variable because higher correlation values mean that the feature is more likely to be useful for prediction. It also helps find multicollinearity, which happens when features that are very similar to each other can add unnecessary information and make the model work worse.



E. Feature Selection

The input features that were chosen were those that had a physical meaning and affected thermal conductivity. This process makes the model work better, cuts down on noise, and makes sure that only important variables are used to make predictions. The main goal of this study is to predict an increase in thermal conductivity, so the target variable (***y***) was set up that way.

F. Data Splitting

We split the dataset into training and testing sets to see how well the model worked. About 80% of the data was used to train the models, and the other 20% was kept for testing. During the splitting process, a fixed random state was used to ensure reproducibility of results.

G. Feature Scaling

Feature scaling was used to make the data standard, so that all features have the same effect on the model and there is no bias because of differences in scale. This step also makes the model converge faster and use less computer power. Using standardization methods based on the mean and standard deviation, the features were made to have the same value.

III. Model

A. Linear Regression

Linear Regression is a basic straightforward supervised machine learning algorithm employed to forecast continuous values by presuming a linear correlation between input features and the target variable. It fits a straight line by using methods like least squares to make the difference between the predicted and actual values as small as possible. It's easy to understand and works quickly, but it might have trouble with patterns that aren't straight lines.

$$\hat{y} = \beta_0 + \sum_{i=1}^{n} \beta_i x_i$$

We used Linear Regression as a baseline model in this project to see how well a simple linear approach can predict an increase in thermal conductivity and to see how it compares to more advanced models.

B. Support Vector Regression(SVR)

Support Vector Regression (SVR) is a Support Vector Machine that is used for regression tasks. SVR doesn't directly try to make mistakes smaller. Instead, it tries to fit a function within a tolerance margin (ε), which means that errors that are within the margin are not counted. It can deal with non-linear relationships thanks to kernel functions (like RBF), which makes it great for large datasets. SVR handles high-dimensional data well, but it can be costly to run and needs careful tuning of its settings. This project used SVR because the link between thermophysical properties and thermal conductivity enhancement is very non-linear, and SVR is good at finding these kinds of patterns.

$$\hat{y} = \sum_{i=1}^{n} (\alpha_i - \alpha_i^*) K(x_i, x) + b$$

C. Random Forest Regression

The Random Forest Regression model is a type of supervised ensemble learning that makes a lot of decision trees and then combines their results to make predictions more accurate. For each tree in the forest, the model randomly picks both observations and predictor variables from the training data set. The Random Forest Regression model makes a prediction by averaging the predictions of all the decision trees.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_t(x)$$

This project used Random Forest to make predictions more accurate and cut down on overfitting by using more than one decision tree and finding complex feature interactions in the data set.

D. Extreme Gradient Boosting(*XGBoost)*

XGBoost (Extreme Gradient Boosting) is a way to boost gradient outputs by building trees one after the other, with each tree fixing mistakes made by the trees before it. Instead, it uses regularization methods, parallel processing, and smart handling of missing values to get high accuracy and speed. XGBoost is a popular choice for competitions and real-world use, especially when working with structured or tabular data.

$$\hat{y} = \sum_{t=1}^{T} f_t(x)$$

XGBoost was used in this project because it works well with structured data and can find complex nonlinear relationships in thermal conductivity enhancement. It also has high predictive performance and efficiency.

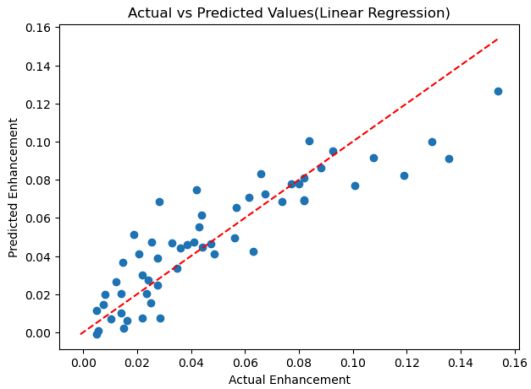
IV. Result and Discussion

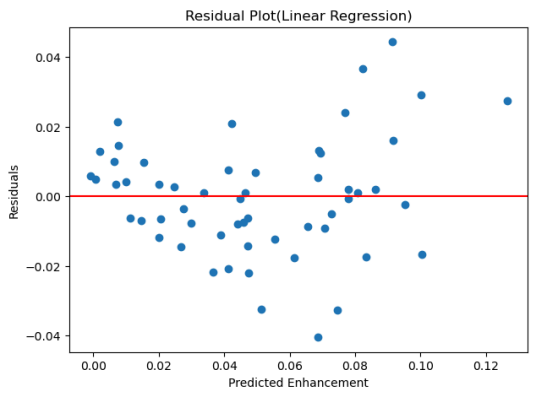
From the study and by training the dataset on multiple machine learning models including Linear Regression(LR), Support Vector Regression(SVR), Random Forest(RF) and eXtreme Gradient Boosting(XGBoost), we predicted the target variable. The performance of all the machine learning models were evaluated using the regression metrics that are the Coefficient of determination(R^2 Score), Mean Absolute Error(MAE) and Root Mean Square Error(RMSE). Along with these numbers, we also used visual evaluation methods like Actual vs. Predicted plots and residual plots to check how well the model worked and how errors were spread out. This gave us a better idea of how well the model worked.

Model	R^2 Score	MAE	RMSE
Linear Regression	0.7934	0.01265	0.01646
SVR	0.9230	0.00786	0.01005
Random Forest	0.9081	0.00734	0.01098
XGBoost	0.9095	0.00727	0.01089

A. Linear Regression

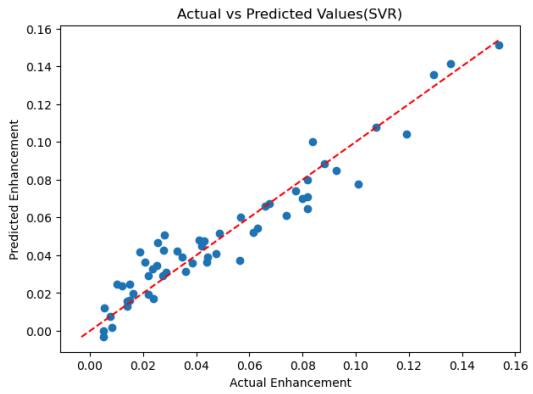
The Linear Regression (LR) model has achieved a R^2 Score of **0.7934,** which tells us it is a moderate fit to the dataset. The values of Mean Absolute Error and the Root Mean Squared Error are relatively high, suggesting that it is difficult to capture the non-linear relations present in the data with this model, and this is highly expected as linear regression gives us a linear relationship between the features and the target variable. In the Actual vs Predicted Plot, we can see a noticeable deviation of data points from the reference line(dotted red line) at the extreme values, which tells us about the underfitting of the data. Furthermore, the Residual Plot tells us about a clear pattern rather than a random distribution of data, that the model fails to capture the underlying data complexity and that's why it introduces the errors.

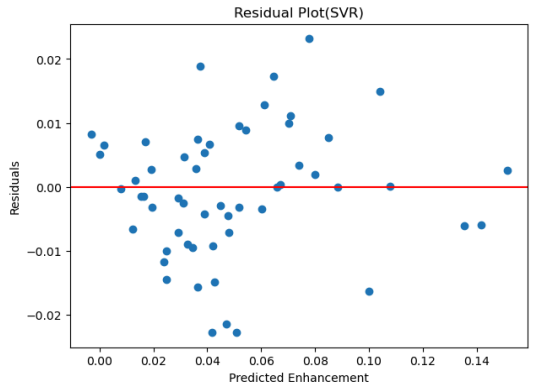




B. Support Vector Machine(SVM)

The Support Vector Machine(SVM) Mode shows us a significantly improved performance, and it has achieved the highest R^2 Score of **0.9230**, which tells us a strong correlation or a close fit. The values of both Mean Absolute Error and the Root Mean Squared Error have reduced, indicating that the SVR Model is effectively capturing the non-linear patterns of the dataset too using the kernel functions and, and it tells us about its robustness in handling the complex relationships. The Actual vs Predicted Plot tells us that mostly the data points are closely aligned with the regression line, indicating a high accuracy and the residual plot shows the residuals randomly scattered around the zero line with relatively low variance which tells us that the model is well-fitted, unbiased and has good generalization capability.the underlying data complexity and that's why it introduces the errors.

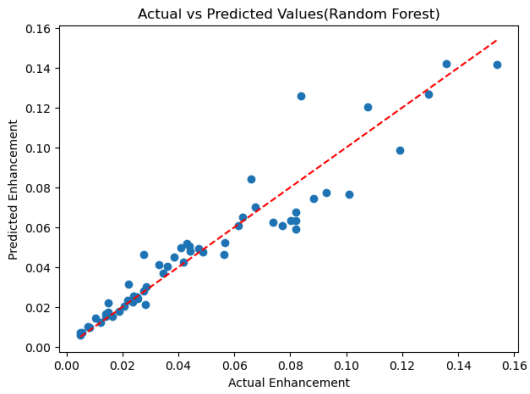


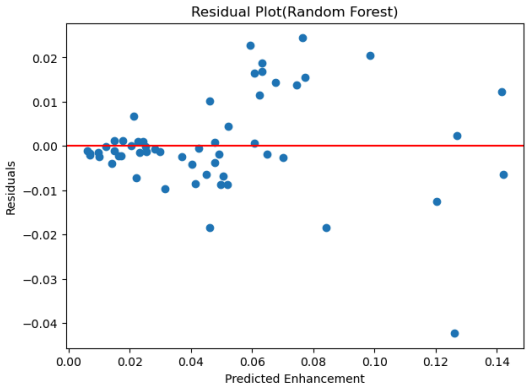


C. Random Forest Model

Random Forest Model also gave us a good performance, having an R2 Score of **0.9081**. Being an ensemble method, it reduces the problem of overfitting and improves the accuracy by combining multiple decision trees. The Mean Absolute Error and the Root Mean Squared Error are relatively low, which suggests consistent prediction performance. The slightly higher RMSE compared to SVR and XGBoost tells us the model performs with good consistency and marginally larger deviations in some predictions. The Actual vs Predicted Plot shows that a majority of data points are aligned near the original line, but a slight spread is also visible, indicating a slight variance. The residual plot shows residual dispersed moderately

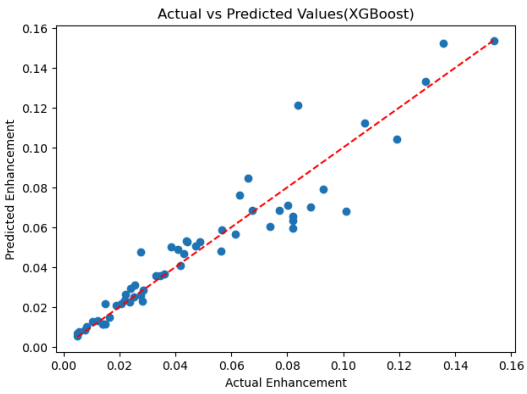
around the zero line, indicating a stable performance but slightly less precise than SVR.

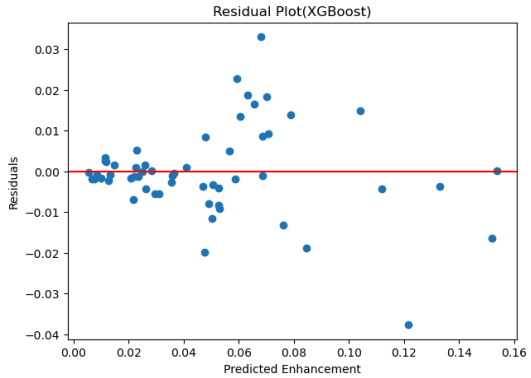




D. *XGBoost Model*

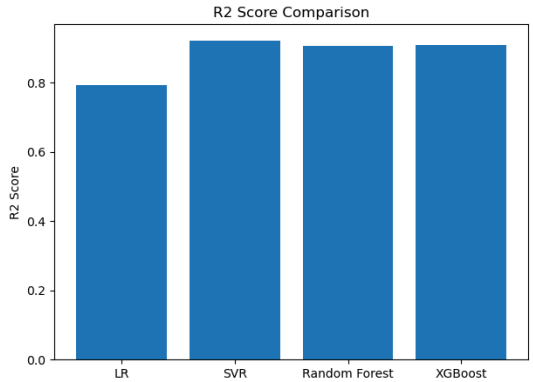
XGBoost model has achieved an R2 Score value of 0.9095 making it one of the top-performing models. With the help of its boosting mechanism, XGBoost lowers the prediction error of the prior prediction, making it a reliable performer. Although its RMSE (0.01089) is a bit higher than SVR, it achieves the lowest MAE (0.00727). This shows it has more consistent average prediction accuracy across the dataset. The Actual vs Predicted plot shows that the points are closely grouped along the regression line. This indicates strong predictive ability. The residual plot supports this as well. It displays a scattered and random distribution of residuals around zero. This implies low error and solid generalization performance.

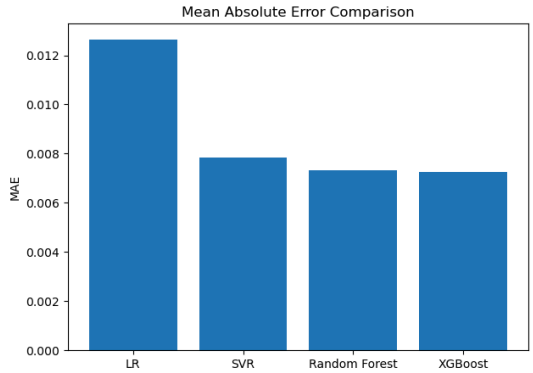


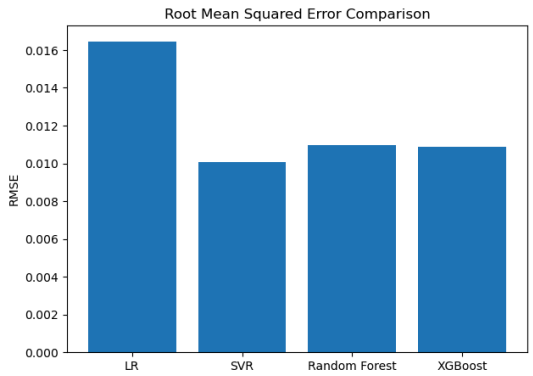


V. Comparative Analysis of ML-Models

The findings indicate that non-linear and ensemble models surpass Linear Regression for this dataset. SVR had the best overall performance of all the models because it had the highest R^2 and the lowest RMSE, which means it was the most accurate at making predictions. XGBoost also did well, giving the lowest MAE and making predictions that were always correct. Random Forest, on the other hand, showed stable results. SVR is the best model for accuracy overall, but XGBoost is a strong second choice.





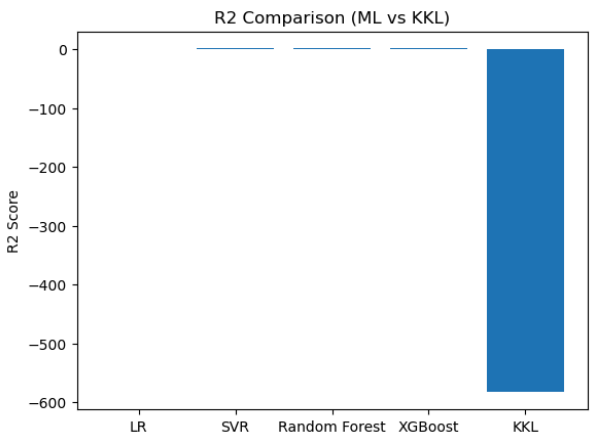


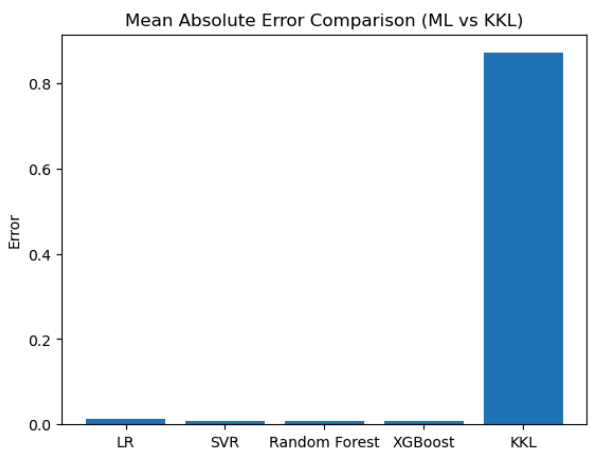
VI. Comparative Analysis of KKL-Model

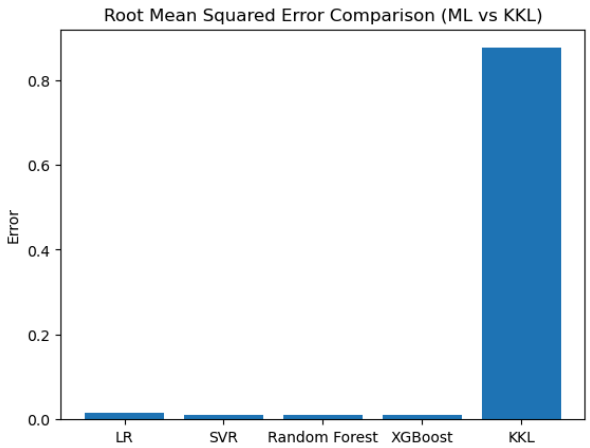
The comparative analysis of the traditional KKL model and machine learning approaches demonstrates a notable disparity in predictive performance. The KKL model does not fit the data very well, as shown by the very low R^2 value **(-582.85)** and the very high MAE **(0.871)** and RMSE **(0.875)**. A negative R^2 of this size means that the model does much worse than a simple mean-based prediction. This shows that it can't pick up on the underlying data trends.

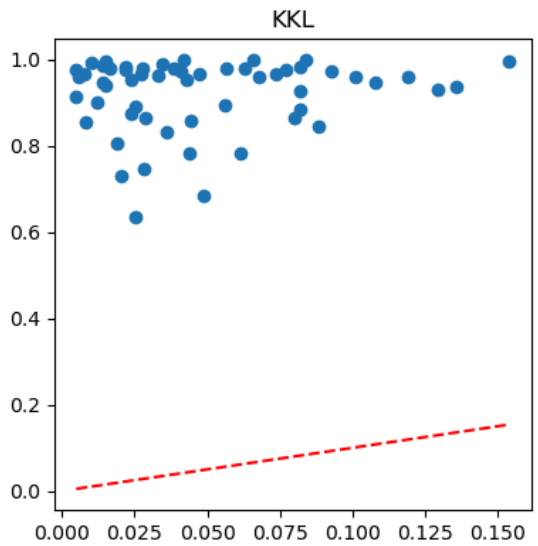
Model	R^2 Score	MAE	RMSE
Linear Regression	0.7934	0.01265	0.01646
SVR	**0.9230**	0.00786	**0.01005**
Random Forest	0.9081	0.00734	0.01098
XGBoost	0.9095	**0.00727**	0.01089
KKL Model	-582.857807	0.871046	0.875348

On the other hand, all of the machine learning models are very good at making predictions, with R^2 values above **0.79** and much lower error metrics. SVR and XGBoost stand out as the best because they can model complex nonlinear relationships. The graphical analysis backs this up even more, showing that ML models make predictions that are very close to the real values and that the residuals are randomly spread out around zero. The KKL model, on the other hand, has a plot of Actual vs. Predicted that is completely out of line, and its residuals are big and not random, which means that there is a systematic error and the model doesn't generalise well.









The KKL model doesn't work well because it is based on theoretical ideas and makes assumptions about how variables relate to each other. Models like these often can't work with real-world datasets that have noise, non-linearity, and complicated feature interactions. Machine learning models, on the other hand, learn patterns directly from the dataset, which helps them find more complex relationships and changes.

VII. Conclusion

This study established an extensive machine learning framework for forecasting the enhancement of thermal conductivity in nanofluids, utilizing essential thermophysical parameters including temperature, particle size, volume fraction, and conductivity values. The dataset was methodically preprocessed, and domain-specific feature engineering was executed, incorporating the conductivity ratio (k_p/k_m) and defining thermal conductivity enhancement as the target variable. These improvements made it easier for the models to understand the physical laws that control how heat moves. We used standard evaluation metrics like R^2 score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) to compare Linear Regression, Support Vector Regression (SVR), Random Forest, and XGBoost.

SVR was the best at making predictions among the models. It had the highest R^2 score and the lowest RMSE, which showed that it was good at modelling complex nonlinear interactions. Ensemble methods like Random Forest and XGBoost also showed strong and steady performance, which supports the idea that there are nonlinear dependencies in the dataset.

On the other hand, the traditional Koo–Kleinstreuer–Li (KKL) theoretical model did not do well at all. It had a very low R^2 value and big prediction errors. This shows that traditional analytical models can't fully capture how complex and variable real-world nanofluid systems are. The results show that using machine learning based on data is a more accurate and dependable way to model how thermal conductivity improves.

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